

GA-Driven Recurrent Modeling for Indonesian Banking Stocks Forecasting: LSTM and Bi-LSTM with Explainable AI and Statistical Validation

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Abstract—This paper studies daily closing-price forecasting for major Indonesian banking stocks using recurrent sequence models, macroeconomic regressors, and statistical validation. The implemented ablation compares LSTM and BiLSTM under two feature settings (`Without_Macro` and `With_Macro`) and two optimization modes (default vs budgeted tuning block), resulting in eight experimental cases per bank. To improve rigor, we run repeated multi-seed evaluations (five seeds) across seven banks and report RMSE, MAE, MAPE, and MSE. We further apply Wilcoxon signed-rank and Diebold-Mariano tests to verify whether performance gains are robust rather than stochastic artifacts. Empirically, BiLSTM-based configurations dominate most banks, with best-case RMSE improvements over the baseline ranging from 4.91% to 35.66% for six out of seven banks. Diebold-Mariano results indicate significant forecast-error reduction in those banks, while Wilcoxon significance is limited by the small number of seeds. The study provides a reproducible benchmark and a statistically grounded foundation for testing whether macroeconomic feature engineering consistently improves forecasting.

Keywords—stock forecasting, LSTM, BiLSTM, macroeconomic features, genetic algorithm, Wilcoxon test, Diebold-Mariano test

I. INTRODUCTION

Forecasting bank stock prices remains challenging due to non-stationarity, regime shifts, and sensitivity to macroeconomic conditions. In emerging markets such as Indonesia, banking-sector prices are influenced not only by internal price dynamics but also by external drivers including interest rates, inflation, exchange rates, and broad market movements.

Recent work in financial time series forecasting has adopted deep learning models, especially recurrent networks, to capture temporal dependencies. However, many studies still suffer from limited ablation depth and weak statistical validation. These gaps reduce confidence in whether reported improvements generalize across assets and random seeds.

This study develops a reproducible pipeline for Indonesian banking stocks that combines sequence modeling, macroeconomic regressors, repeated-seed evaluation, formal significance testing, and explainability analysis. A central question is whether feature engineering with macroeconomic variables (`With_Macro`) provides consistent gains over the baseline regime (`Without_Macro`), as often suggested in exogenous-variable forecasting literature [1], [2]. In addition, we evaluate whether model behavior remains economically plausible under Explainable AI (XAI) inspection using SHAP-based attributions [3]. The objective

is to provide a strong foundational benchmark that can be extended to more advanced architectures.

The main contributions are as follows:

- A structured ablation design over model type (LSTM vs BiLSTM), feature regime (price-only vs price+macro), and optimization mode (off vs on).
- A multi-bank, multi-seed protocol for robust performance estimation and variance analysis.
- Statistical validation using Wilcoxon signed-rank and Diebold-Mariano tests to assess significance of forecasting improvements.
- An XAI layer based on SHAP to quantify global feature relevance and compare explanation patterns across banks, feature regimes, and optimization settings.

The remainder of this paper is organized as follows. Section II reviews related literature. Section III describes the dataset and preprocessing. Section IV presents the proposed methodology. Section V details the experimental setup. Section VI discusses the results and significance tests. Section VII concludes and outlines future directions.

II. RELATED WORKS

Literature on financial time-series forecasting is dominated by recurrent models and newer transformer-style architectures. LSTM-type models remain widely used in stock and index forecasting because they can capture nonlinear temporal dependence with manageable model complexity. Recent IEEE studies on stock/index prediction continue to report competitive recurrent variants, including BiLSTM-based designs [4], [5].

Another key direction is forecasting with exogenous variables. Recent studies show that incorporating external drivers can improve prediction accuracy when those signals are informative and temporally aligned [1], [2], [6]–[8]. This motivates our explicit comparison between `Without_Macro` and `With_Macro` settings.

Explainability is also becoming important in financial AI because predictive accuracy alone is insufficient for risk-sensitive decisions. SHAP provides a model-agnostic additive attribution framework and has been used in stock-market factor analysis to interpret model drivers [3], [9]–[11]. Recent IEEE studies also emphasize XAI-oriented analysis for financial forecasting workflows, including LRP- and dashboard-based explainability settings [12]–[15]. Compared with black-box reporting, SHAP-based summaries can reveal whether the dominant drivers are stable across assets, seeds, and modeling regimes.

Beyond architecture and features, methodological rigor remains a common weakness in applied financial forecasting papers. Many studies report single-run metrics without repeated-seed robustness checks or formal significance testing. This paper addresses that gap by combining controlled ablation, multi-seed evaluation, and paired statistical tests (Wilcoxon and Diebold-Mariano) in one reproducible pipeline for Indonesian banking stocks.

III. DATASET AND PREPROCESSING

A. Dataset Description

This study focuses on seven Indonesian banking stocks: BBKA, BBNI, BBRI, BBTN, BDMN, BMRI, and BNGA. Daily market data include price and valuation-related variables, while macroeconomic variables include inflation-related indicators, policy rate, exchange rate, and market index features.

The target variable is the daily closing price. The analysis period follows a chronological split to avoid data leakage.

B. Feature Selection

The baseline feature set (*Without_Macro*) includes market micro and technical variables available in the bank-level dataset. The macro-augmented regime (*With_Macro*) extends this set with external variables:

- inflation rate and CPI-derived indicators,
- BI policy rate,
- USD/IDR exchange rate,
- IHSG index level and return proxy.

The final feature schema is fixed per regime before model training.

C. XAI-Oriented Feature Schema

To support consistent SHAP analysis across all experiments, feature names are kept identical across banks and seeds, and the same temporal windowing process is used in every run. The *Without_Macro* regime uses 10 market/valuation features, while *With_Macro* adds 6 external variables (total 16 features). This fixed schema allows direct aggregation of feature attribution statistics across the full ablation matrix.

D. Data Preprocessing

We apply chronological sorting, forward-fill for missing observations, and standard scaling for both features and target during model training. Sequence windows are then constructed with a fixed lookback length.

Time split protocol:

- Training: 2014-01-02 to 2019-12-31
- Validation: 2020-01-02 to 2021-12-31
- Testing: 2022-01-02 to 2024-12-30

This split reflects strict temporal validation and supports fair out-of-sample evaluation.

IV. PROPOSED METHODOLOGY

A. Overall Framework

The pipeline consists of five stages: data loading and scaling, sequence generation, model training, out-of-sample forecasting, and interpretability analysis. We evaluate multiple design factors using controlled ablation to isolate contribution from architecture, macro features, and optimization. Figure 1 summarizes the end-to-end workflow used in this study.

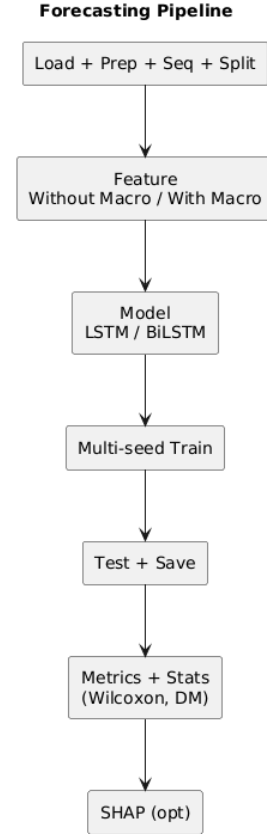


Fig. 1. Compact vertical pipeline overview used for the ablation and evaluation workflow.

B. Forecasting Models

1) *LSTM and BiLSTM*: We use a one-layer recurrent encoder followed by a fully connected prediction head. Two architecture variants are compared:

- LSTM: unidirectional temporal encoder.
- BiLSTM: bidirectional temporal encoder with doubled hidden-state output dimension.

C. Budgeted Hyperparameter Optimization Block

Hyperparameter tuning is implemented as a fixed-budget search process using validation loss as the objective. In this paper, configurations tagged as “_ga” represent the optimization-enabled setting under identical search budget across comparable cases. This design is used to test whether a constrained tuning budget improves accuracy and stability versus default settings.

TABLE I
BEST-PERFORMING CASE PER BANK (MEAN $\pm$ STD OVER 5 SEEDS).

Bank	Best case	RMSE	MAE	MAPE (%)
BBCA	<code>bilstm_without_macro</code>	1881.53 $\pm$ 199.49	1721.55	18.36
BBNI	<code>bilstm_without_macro</code>	335.85 $\pm$ 57.51	252.66	4.90
BBRI	<code>bilstm_without_macro</code>	525.70 $\pm$ 123.24	435.44	8.34
BBTN	<code>bilstm_without_macro_ga</code>	51.44 $\pm$ 16.04	43.48	3.23
BDMN	<code>lstm_without_macro</code>	120.72 $\pm$ 21.92	98.32	3.74
BMRI	<code>bilstm_with_macro_ga</code>	1388.71 $\pm$ 257.96	1165.79	19.09
BNGA	<code>bilstm_without_macro</code>	244.14 $\pm$ 28.12	192.77	11.23

D. Statistical Validation

Beyond average error reporting, we apply:

- Wilcoxon signed-rank test on seed-level paired RMSE,
- Diebold-Mariano test on aligned forecast residuals.

These tests assess whether gains over baseline are statistically significant.

V. EXPERIMENTAL SETUP

A. Evaluation Metrics

Primary metrics are MSE, RMSE, MAE, and MAPE on the test partition. We also report seed-wise variability (mean and standard deviation) to characterize stability.

B. Ablation Matrix

The main ablation factors are:

- Model family: LSTM and BiLSTM
- Features: `Without_Macro` vs `With_Macro`
- Optimization mode: default vs optimization-enabled (“_ga”)

This yields eight recurrent-model cases per bank for controlled comparison.

C. Multi-Seed Protocol

Each configuration is executed with five seeds (42, 52, 62, 72, 82). The complete run table stores configuration meta-data, metrics, and file paths for downstream significance analysis.

D. Implementation Details

Implementation is based on Python and PyTorch. Training uses Adam optimizer, ReduceLROnPlateau scheduling, and early stopping with validation monitoring. Sequence length, forecasting horizon, and tuning budget are controlled via command-line arguments for reproducibility.

E. Cross-Model Fairness Protocol

To maintain fairness, all cases use identical data splits, seed sets, sequence lengths, and optimization budgets. This avoids overstating gains from unequal search effort.

VI. RESULTS AND DISCUSSION

A. Performance Comparison Across Configurations

Table I reports the best case per bank based on mean RMSE across five seeds. BiLSTM-based settings are dominant in six out of seven banks. The best configurations are mostly `bilstm_without_macro`, with two bank-specific exceptions: `bilstm_without_macro_ga` for BBTN and `bilstm_with_macro_ga` for BMRI.

Across all eight ablation cases, the best overall average RMSE is obtained by `bilstm_without_macro_ga`

TABLE II
BEST CASE VS BASELINE (`LSTM_WITHOUT_MACRO`) SIGNIFICANCE SUMMARY.

Bank	Best case	Improve (%)	Wilcoxon p	DM p
BBCA	<code>bilstm_without_macro</code>	8.76	0.4375	2.28×10^{-13}
BBNI	<code>bilstm_without_macro</code>	10.47	0.0625	4.84×10^{-11}
BBRI	<code>bilstm_without_macro</code>	28.23	0.3125	$< 10^{-16}$
BBTN	<code>bilstm_without_macro_ga</code>	35.66	0.3125	2.22×10^{-16}
BDMN	<code>lstm_without_macro</code>	0.00	1.0000	n/a
BMRI	<code>bilstm_with_macro_ga</code>	11.18	0.4375	2.23×10^{-11}
BNGA	<code>bilstm_without_macro</code>	4.91	0.3125	4.78×10^{-5}

TABLE III
OVERALL RANKING ACROSS ALL BANKS (LOWER IS BETTER).

Case	RMSE	MAE	MAPE (%)
<code>bilstm_without_macro_ga</code>	680.83	591.04	10.65
<code>bilstm_without_macro</code>	681.26	589.76	10.74
<code>lstm_without_macro</code>	741.52	644.21	11.44
<code>lstm_without_macro_ga</code>	781.95	688.86	12.66
<code>bilstm_with_macro_ga</code>	806.40	722.52	13.76
<code>bilstm_with_macro</code>	814.04	720.31	13.14
<code>lstm_with_macro</code>	851.89	759.88	13.96
<code>lstm_with_macro_ga</code>	938.15	842.31	17.48

(680.83), followed closely by `bilstm_without_macro` (681.26), while `lstm_without_macro` is third (741.52). In this dataset, moving from `Without_Macro` to `With_Macro` is not uniformly beneficial, and improvements are bank-dependent.

B. Statistical Significance Analysis

Table II compares each bank’s best case against `lstm_without_macro` as baseline. RMSE gains are positive in six banks, ranging from 4.91% to 35.66%.

C. Global Ranking and Regime Matrix

Table III presents an overall ranking across all banks using mean RMSE, MAE, and MAPE. The top two configurations are both BiLSTM-Without_Macro variants, while all With_Macro variants rank below them in the aggregate view.

To directly answer whether macroeconomic feature engineering improves accuracy, Table IV reports the RMSE change from `Without_Macro` to `With_Macro` per bank. Positive values indicate degradation, and negative values indicate improvement.

This evidence suggests that macro variables are useful only in specific regimes/banks rather than universally. This pattern is still compatible with the broader literature: methods designed for exogenous variables can improve forecasting when the external signals are informative and properly aligned [1], [6], and multiple IEEE studies also report gains from macro-integrated forecasting pipelines [2], [7], [8]. In our setting, the net effect is mixed, indicating that feature relevance and temporal alignment dominate over simple feature expansion.

D. XAI Analysis with SHAP

We generated SHAP outputs for all 280 trained checkpoints (7 banks $\times$ 8 cases $\times$ 5 seeds). This produced 3,640

TABLE IV
RMSE CHANGE (%) FROM WITHOUT_MACRO TO WITH_MACRO.

Bank	LSTM delta (%)	BiLSTM delta (%)
BBCA	+20.34	+27.87
BBNI	+30.70	+39.21
BBRI	-1.27	+23.19
BBTN	-5.04	-17.72
BDMN	+202.86	+126.82
BMRI	-1.76	-4.85
BNGA	+13.25	+15.45
Mean	+37.01	+29.99

TABLE V
GLOBAL SHAP SUMMARY (ROBUST AGGREGATE OVER ALL 280 RUNS).

Feature	Mean —SHAP— (cap99)	Top-1 count
Low Price	0.00465	39
Bid Price	0.00428	31
Price Change 1 Day Percent	0.00407	28
High Price	0.00394	25
Moving Average 20 Day	0.00383	25
Price to Book Ratio	0.00383	25
Price Earnings Ratio	0.00365	20
Price to Sales Ratio	0.00362	28

feature-attribution rows over 16 unique features. Using global top-1 frequency, the most frequently dominant drivers are *Low Price* (39 runs), *Bid Price* (31), *Price Change 1 Day Percent* (28), and *Price to Sales Ratio* (28). The pattern indicates that short-horizon market microstructure and valuation factors dominate the model explanations more often than macro indicators.

Because a small number of runs showed extreme SHAP magnitudes (maximum $\approx 1.51 \times 10^{13}$), we also report a robust global ranking using a 99th-percentile cap. Under this robust aggregation, the top features remain stable: *Low Price* (0.00465), *Bid Price* (0.00428), *Price Change 1 Day Percent* (0.00407), *High Price* (0.00394), and *Moving Average 20 Day* (0.00383). Hence, the main interpretation does not depend on a few outlier explanations.

At the bank level, the most frequent top-1 SHAP feature differs across issuers: BBCA and BBNI are led by *Bid Price*, BBRI by *Price Change 1 Day Percent*, BBTN and BMRI by *High Price*, BDMN by *Price to Sales Ratio*, and BNGA by *Low Price*. This heterogeneity supports the earlier finding that forecasting behavior is regime- and bank-specific. In macro-enabled cases, *Price Change 1 Day Percent* and *Low Price* remain dominant, while macro variables (e.g., CPI index) appear as top-1 only in a subset of runs.

E. Discussion and Threats to Validity

Three validity points are important. First, the chronological split is fixed; different split boundaries may change relative ranking. Second, seed-level Wilcoxon tests are underpowered with $n = 5$, which explains why several improvements with very small DM p -values are not mirrored by Wilcoxon significance. Third, optimization benefits are

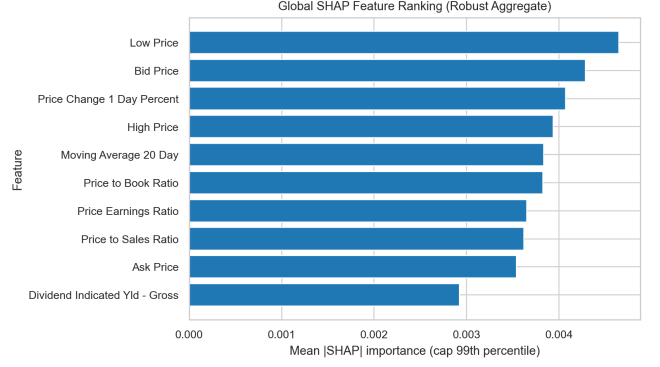


Fig. 2. Global SHAP ranking using robust mean importance (99th-percentile capped).

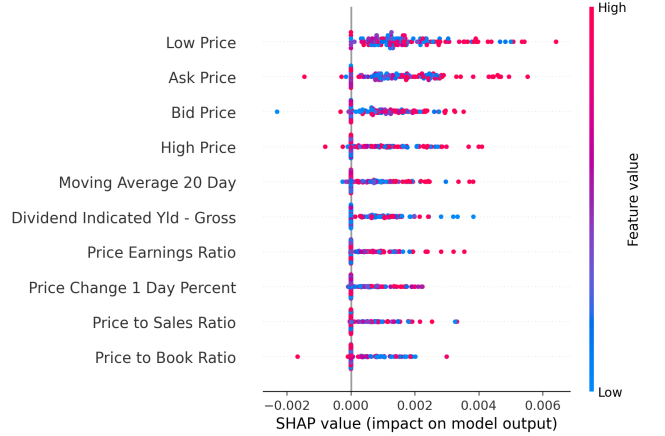


Fig. 3. SHAP beeswarm plot for the best-performing checkpoint (BBTN, BiLSTM_Data_GA, seed 62).

heterogeneous across banks, indicating risk of overfitting when tuning is not controlled.

Future work will extend this benchmark through walk-forward validation, larger seed counts, and broader stress-testing across additional market regimes for stronger temporal robustness.

VII. CONCLUSION

This paper presents a reproducible framework for forecasting Indonesian bank stock prices using LSTM-family models, macroeconomic features, multi-seed evaluation, and statistical significance testing.

Empirical results show that BiLSTM-based settings are generally stronger than LSTM across the evaluated banks, with best-case RMSE gains over the `lstm_data` baseline reaching up to 35.66%. Diebold-Mariano tests confirm significant forecast-error reduction in six banks, while Wilcoxon results are less conclusive due to limited seed count. The evidence also indicates that macro features and optimization are not uniformly beneficial, motivating bank-specific model selection rather than one-size-fits-all deployment.

Immediate future work includes: (i) extending validation to walk-forward protocols, (ii) increasing seed count for stronger nonparametric testing power, and (iii) stress-testing recurrent models under additional market regimes.

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